

INTERNSHIP AT ALTEN

R&D : Urgency in the Digital Twin Smart Green Supply Chain

Measuring declared against real urgency in a supply chain digital twin, so that it can be acted upon

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Cohort 2026 DED

COMPANY

ALTEN Sud-Ouest
Direction de l'Innovation · Lab de Toulouse

PERIOD

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Confidential: YES

TUTORS

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PLAN



1 CONTEXT, PROBLEM, MISSION



2 HOW THE PROJECT WAS RUN



3 DESIGN AND REALISATION



4 RESULTS AND VALIDATION



5 ASSESSMENT AND PERSPECTIVES



THE COMPANY AND THE PROGRAMME

ALTEN · Direction de l'Innovation · Toulouse Lab



1988

FOUNDED



+30

COUNTRIES



11

ALTEN LABS



31 %

R&D ON ENVIRONMENT

SMART GREEN SUPPLY CHAIN: THREE AIMS AT ONCE



LEAN

Cost, delay, excess stock



GREEN

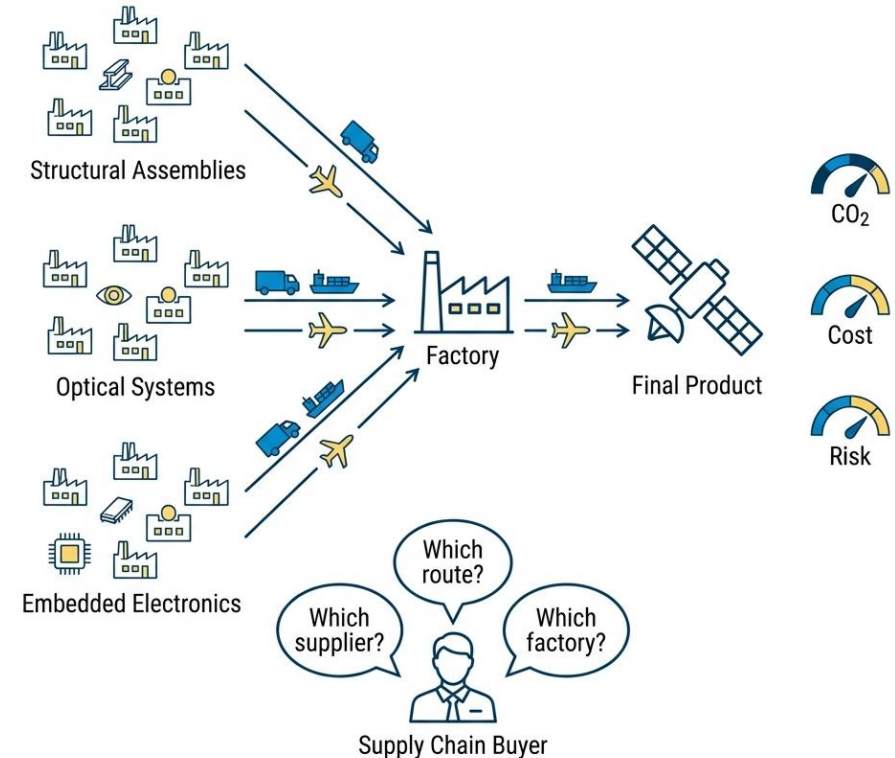
Carbon per transport leg, to GLEC v3.2 and ISO 14083



PREDICTIVE

Anticipation of disruption. **This work sits here.**

DISCO is a serious game and digital twin covering 22 component suppliers. Routing, risk scoring, carbon and transport were all built by earlier internships. None computes urgency.



Players hold one node each, scored on cost, delay and carbon.

3 | 20

ALTEN

THE PROBLEM, STATED



THE TWIN HAS

Computed physical state



THE TWIN NEVER SEES

The human who decides

RESEARCH QUESTION

Can the gap between declared and computed urgency be measured reliably enough, inside a working digital twin, to be acted upon?



It must say **which node is mismanaged**



The adequacy score must be **reliable and explainable.**



Must **beat the raw indicators** a coordinator already has

THE LITERATURE GAP

What is new ?

PUBLISHED RESEARCH



Deadline-driven value. Time-varying, but one task only.



Ripple effect. Propagates, but takes the node score as given.



Rupture indicators. Monitored one at a time, never combined.



Cognitive bias. Quantified in the lab, no operational correction.



AHP, BWM, PROMETHEE. Static, and never compared to a computed value.



Digital twins. Scored on state; their inputs are assumed to arrive.



THE GAP

No system, published or patented, combines a declared urgency, a computed one and their propagated mismatch, then scores the result against recorded outcomes.

THE PATENT RECORD: THREE LINES

IBM. Revenue at risk from measured inputs. No operator opinion anywhere in the claims.

Wipro. Inputs replaced by language. The human narrates; nothing evaluates the narration.

Digital twin control tower. Oriented to automation. The operator receives decisions.



Fifteen years running away from the human declarant.

OBJECTIVES AND INDICATORS

	OBJECTIVE	INDICATOR	DELIVERABLE
01	Define real urgency and an adequacy score	Every term decomposable; aggregation justified	Specification
02	Deliver a working, auditable application	Multi-tenant · every write audited · tests passing	SupplyScore
03	Propagate across tiers, rank by criticality	Real-programme scale, interactive time budget	Propagation services
04	Establish whether the signal predicts outcomes	A stated outcome definition, and a trivial baseline	Pre-registered protocol
05	Capitalise the research for the programme	Every number reproducible from raw artefacts	BST · scripts
06	Observe warehouse capacity instead of asking	A physical result, scored on the operational quantity	Segmentation model

THE WEEK

V-cycle for the model · Agile and Test and Learn for the week

MON Planning The week's objective	TUE Scrum Blockages surface in a day	WED Programme director Scientific decisions	THU Scrum Course correction	FRI Demonstration Shown
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TEST AND LEARN MEANS SOMETHING ONLY IF THINGS WERE ABANDONED



The first urgency model, and why it could not fit



A single time-driven curve: urgency rose only as a deadline approached



It could not represent a node **on time but out of capacity**



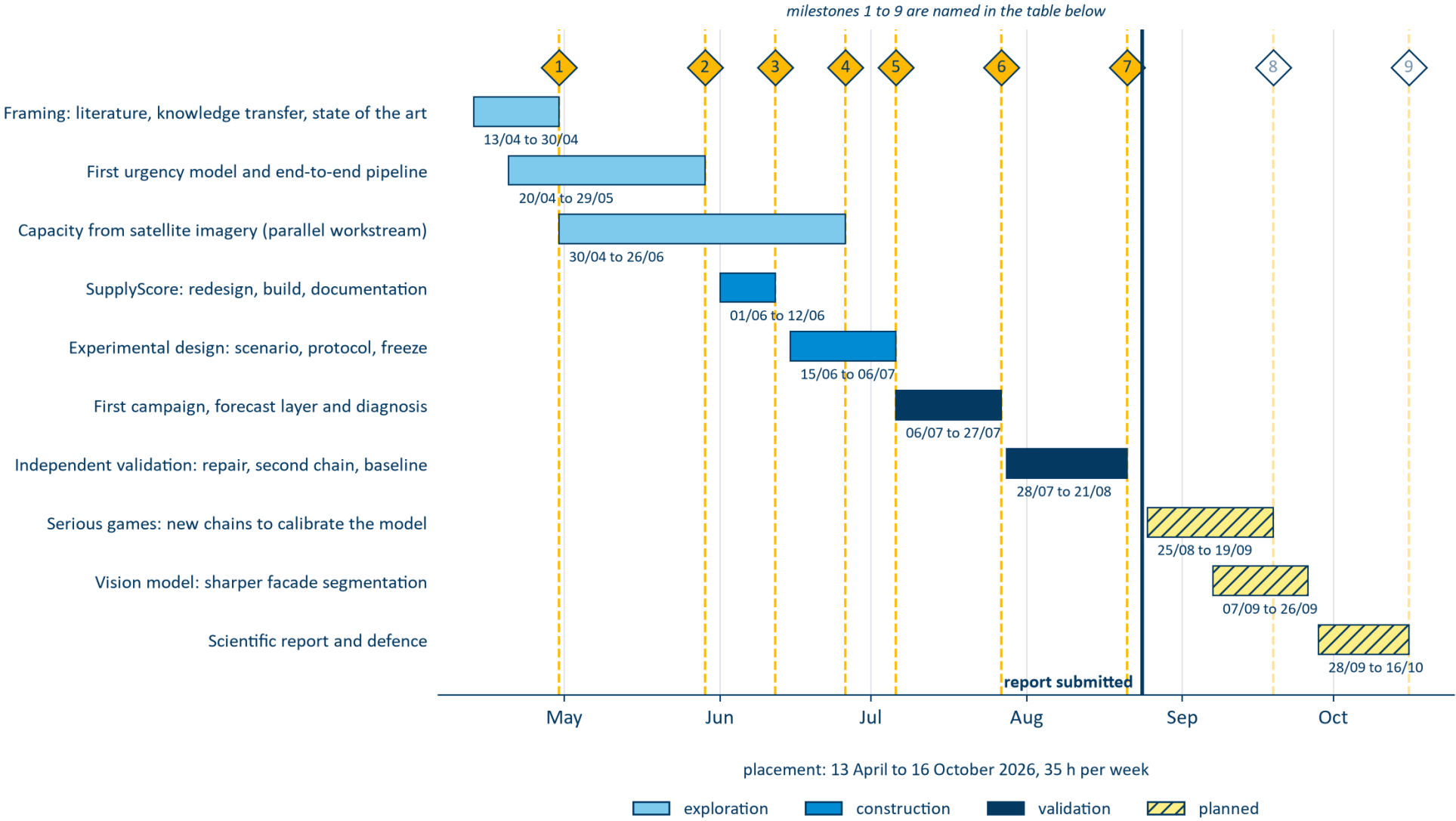
It discarded indicators the twin already held



Dropped in April for multi-block aggregation. Cost: three weeks

PHASES AND MILESTONES

10 phases · 9 milestones



THE MISSION

A working application, two independently elicited signals, and the score that compares them



DECLARED U_D

AHP, FUZZY BWM AND PROMETHEE II

Operational impact

Time window

Downstream dependencies

Recoverability

No declared counterpart



COMPUTED U_R

WEIGHTED PROBABILISTIC OR

Performance and capacity

Time, against the next deadline

Severity propagated across tiers

Risk, through its recovery term

Cost and carbon, arbitration only



ADEQUACY SCORE

The distance between the two signals, on a **0–100** scale.



False urgency · declared > state · **penalty ×1**



Hidden risk · state > declared · **penalty ×2**

Every term stays recoverable, so the operator sees what moved the answer.

REAL URGENCY

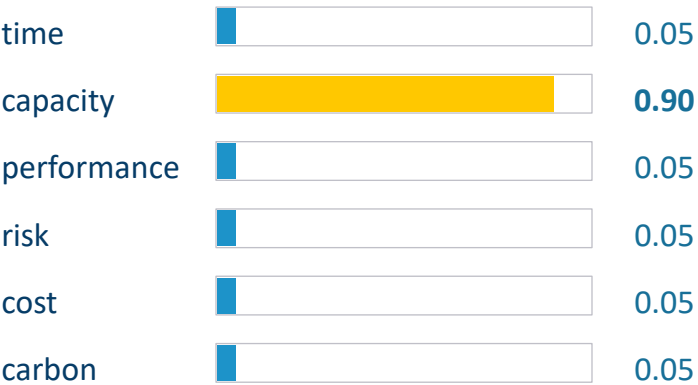
Non-compensatory aggregation score

A node is in trouble if **any one** block fires. A healthy block cannot offset a failed one.

$$U_r = 1 - \prod_m (1 - u_m)^{\omega_m}$$

NODE A

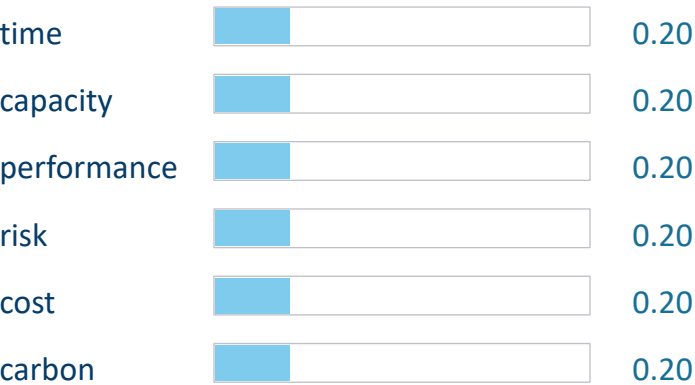
one block has failed



average	0.19 → 2nd
probabilistic OR	0.92 → 1st

NODE B

nothing has failed



average	0.20 → 1st
probabilistic OR	0.74 → 2nd

The average sends the coordinator to **Node B** , where nothing broke. The OR sends them to **Node A** , where capacity did. **Same six numbers, opposite ranking.**

FOUR CAUSES · TWO CONSTRAINTS



$\omega = 0$ means “nothing is known”, kept distinct from $u = 0$, “the block is healthy”.

DECLARED URGENCY

Elicited from the operator, gated on consistency, then propagated downstream

Comparaisons par paires (AHP)
Curseur à gauche : le second critère domine ; à droite : le premier domine ; au centre : importance égale.

« Impact opérationnel » vs « Fenêtre temporelle »

-8

-4

0

+4

+8

Importance égale

0

« Impact opérationnel » vs « Dépendances aval »

-8

-4

0

+4

+8

Importance égale

0

« Impact opérationnel » vs « Récupérabilité »

-8

-4

0

+4

+8

Importance égale

0

« Fenêtre temporelle » vs « Dépendances aval »

-8

-4

0

+4

+8

Importance égale

0

« Fenêtre temporelle » vs « Récupérabilité »

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Importance égale

0

« Dépendances aval » vs « Récupérabilité »

-8

-4

0

+4

+8

Importance égale

0

Interface terms

Comparaisons par paires (AHP)
Pairwise comparisons (AHP)

Impact operationnel
Operational impact

Fenetre temporelle
Time window

Dependances aval
Downstream dependencies

Recuperabilite
Recoverability

Importance egale
Equal importance

Curseur a gauche / a droite
Slider left / right: which side dominates

The application is operated by a French research programme, so its interface is French.

FOUR CRITERIA, EACH WITH A COMPUTED COUNTERPART

Operational impact	Time window
Downstream dependencies	Recoverability



THE CONSISTENCY GATE

A weight vector whose consistency ratio exceeds the threshold is **refused** . The operator is sent back to the most contradictory of their own comparisons.

Without it, an incoherent judgment would propagate as though it were a considered one.



Pairwise comparisons, not a free-text estimate



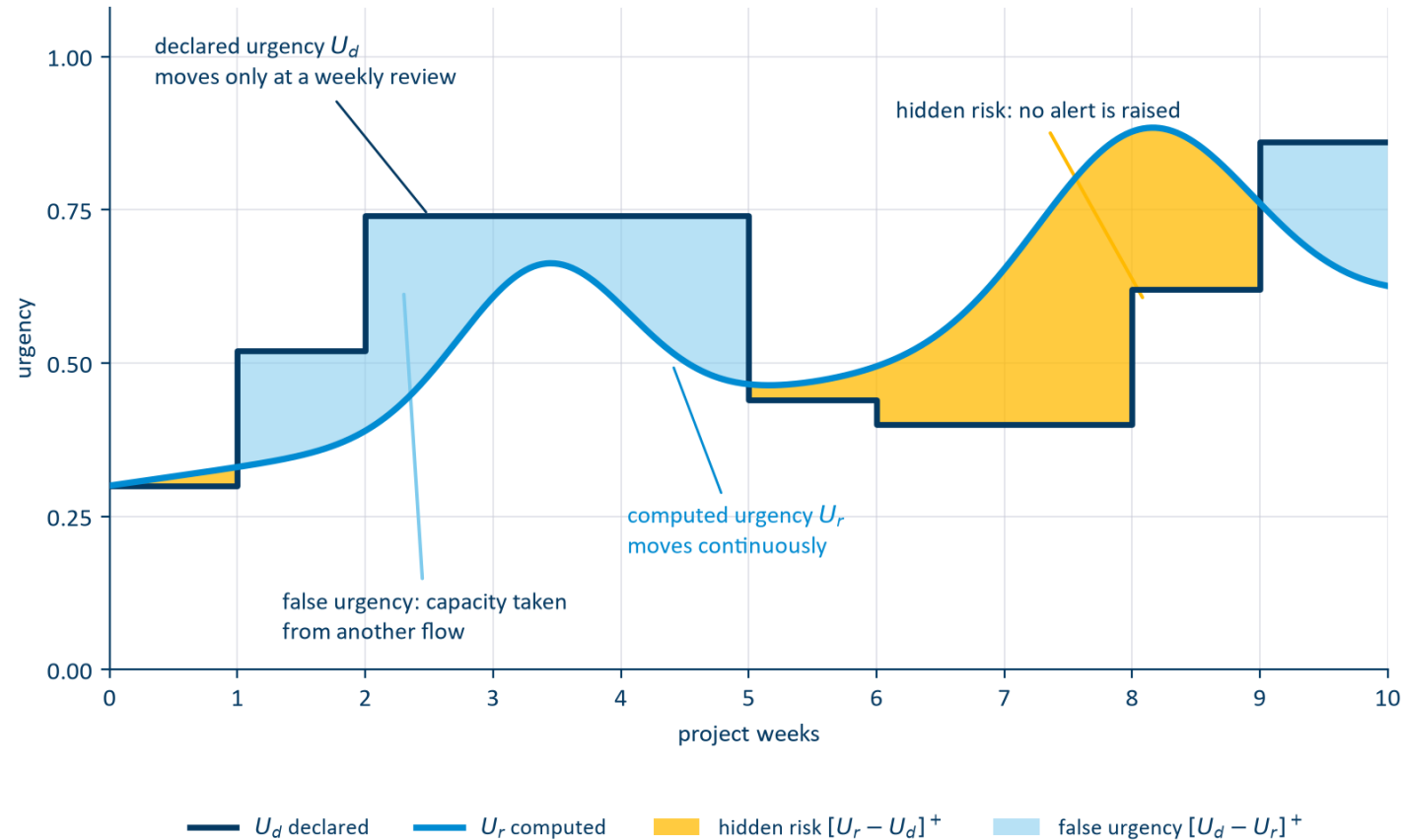
AHP · fuzzy BWM · PROMETHEE II



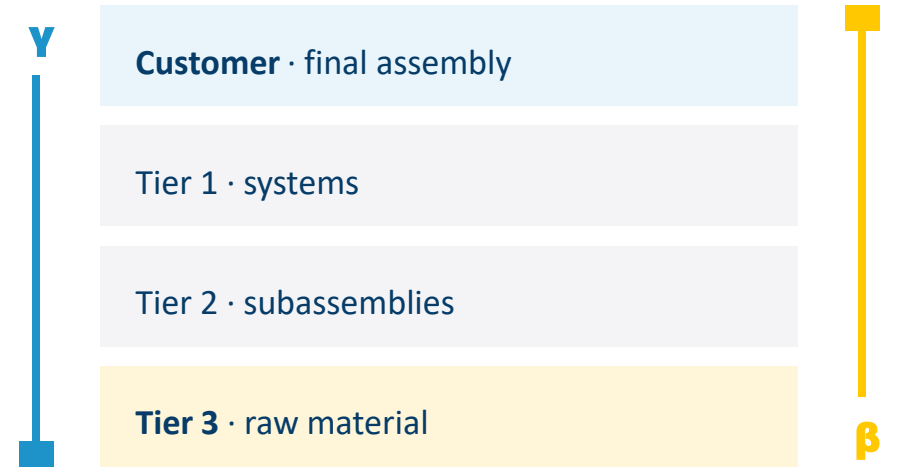
Descends from the customer towards suppliers, attenuated

THE ADEQUACY

Elicited separately, propagated in opposite directions



PROPAGATION ACROSS THE NETWORK



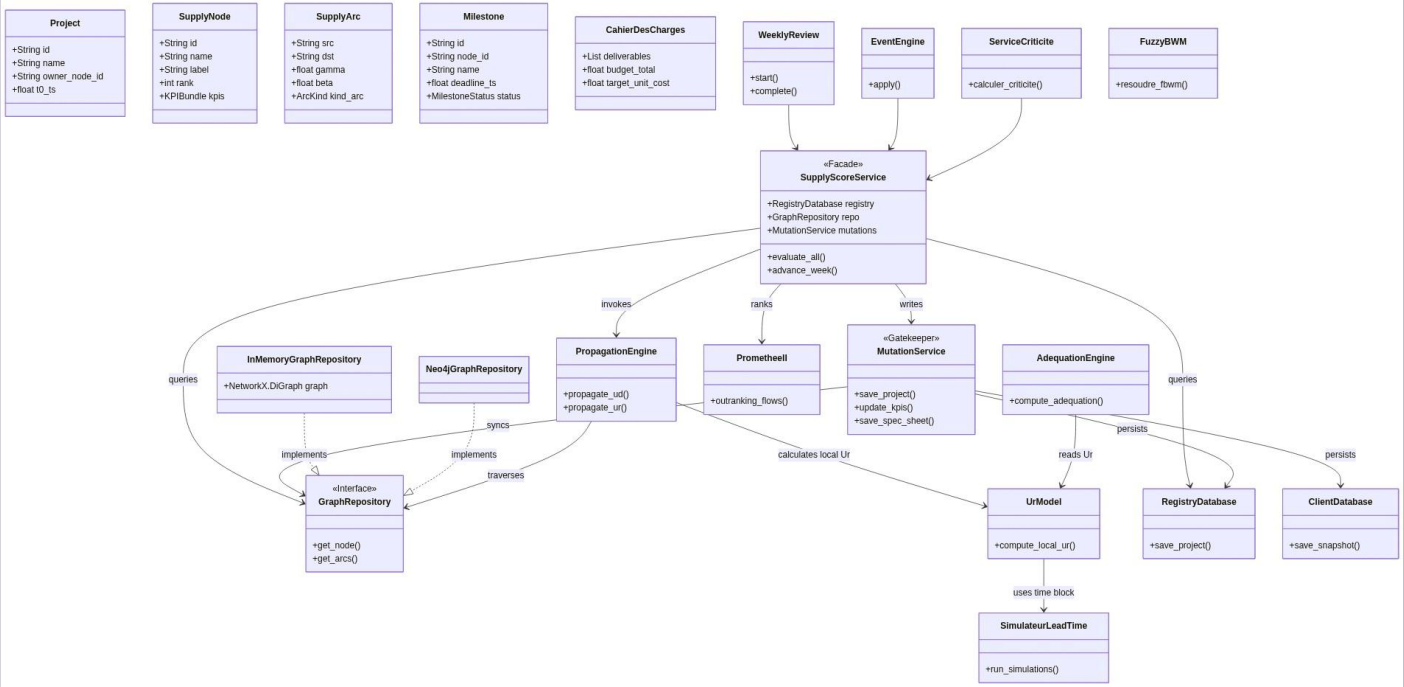
γ attenuates the declaration as it descends: demand pressure is what a customer transmits

β amplifies computed risk as it ascends: physical risk is what a supplier transmits

SUPPLYSCORE: ARCHITECTURE

The class model, and the pages it drives

THE CLASS MODEL



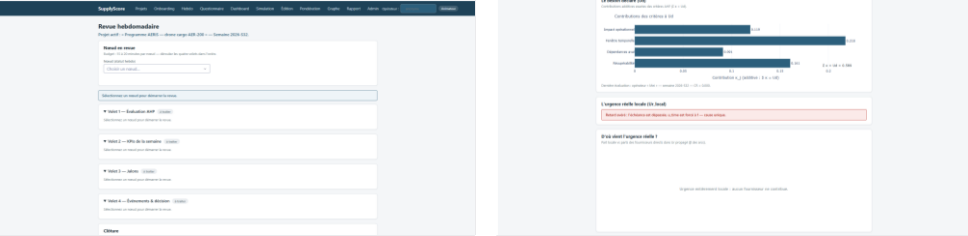
Facade the only entry point

Mutation service the audit trail is architectural

One DB per node isolation by storage layout

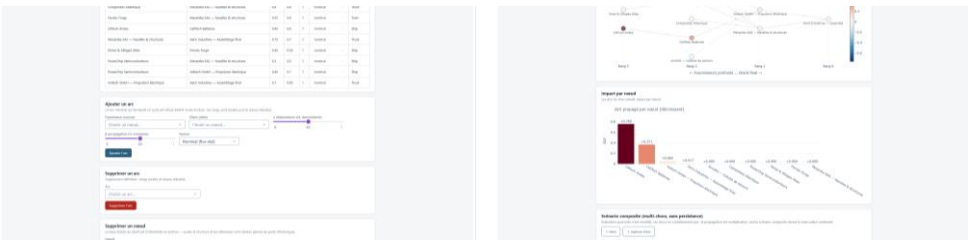
Every arrow into the data crosses the facade, then the mutation service. Two graph repositories sit behind one interface, so the network can be held in memory or in Neo4j without the domain knowing which.

THE APPLICATION, AS THE OPERATOR MEETS IT



Weekly review · one signed transaction

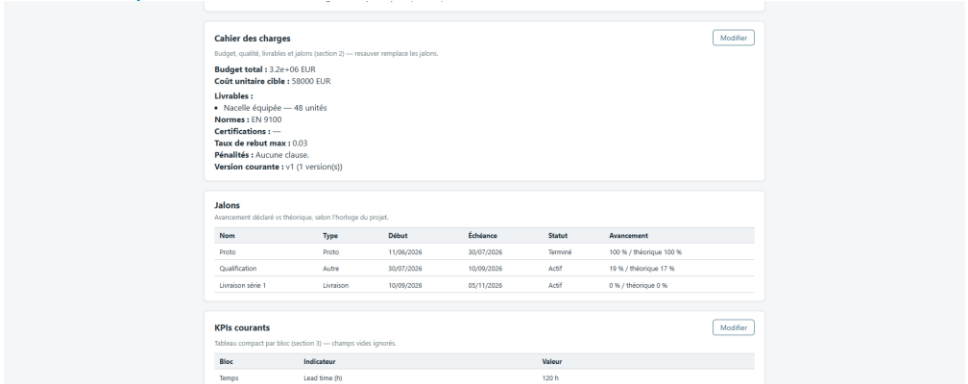
Explainability · term by term



Network · generated per project

Criticality · each node knocked out

A NODE, AND WHAT IT CARRIES



RESULT 1: EARLY WARNING

Nodes ranked by hidden risk above 0.10, against the nodes that later missed

1.00

PRECISION, TURNS 0-5

3 / 4

FUTURE FAILURES FLAGGED

0

FALSE ALARMS IN SIX TURNS

8

TURNS OF ADVANCE WARNING

T0 – T5 · signal sharp

T6 – T18 · crisis public, the gap closes

Precision 1.00 · recall 0.75 · 0 false alarms



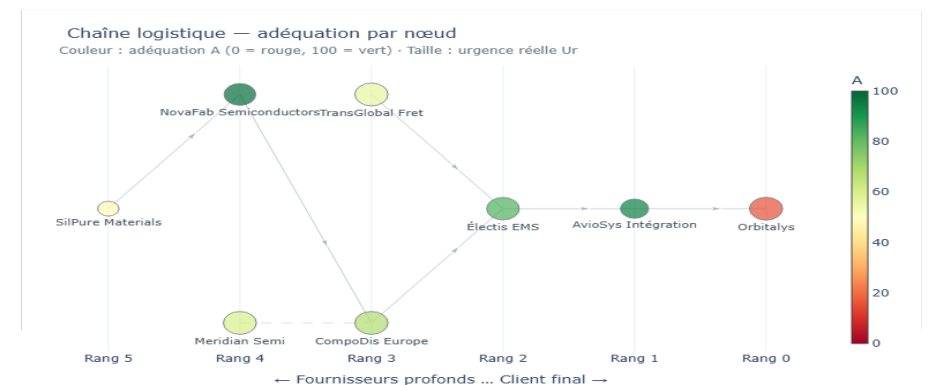
The worst-scoring node is the satellite prime. Flagged from turn 0, it misses at turn 8: **eight turns of warning, while its own manager declares an urgency of 0.001.**



The flag comes from an **optimistic declaration against a computed state that is not**. From turn 6 precision falls to 0.50 or below: the crisis becomes public, everyone raises their declaration, and a difference closes mechanically.

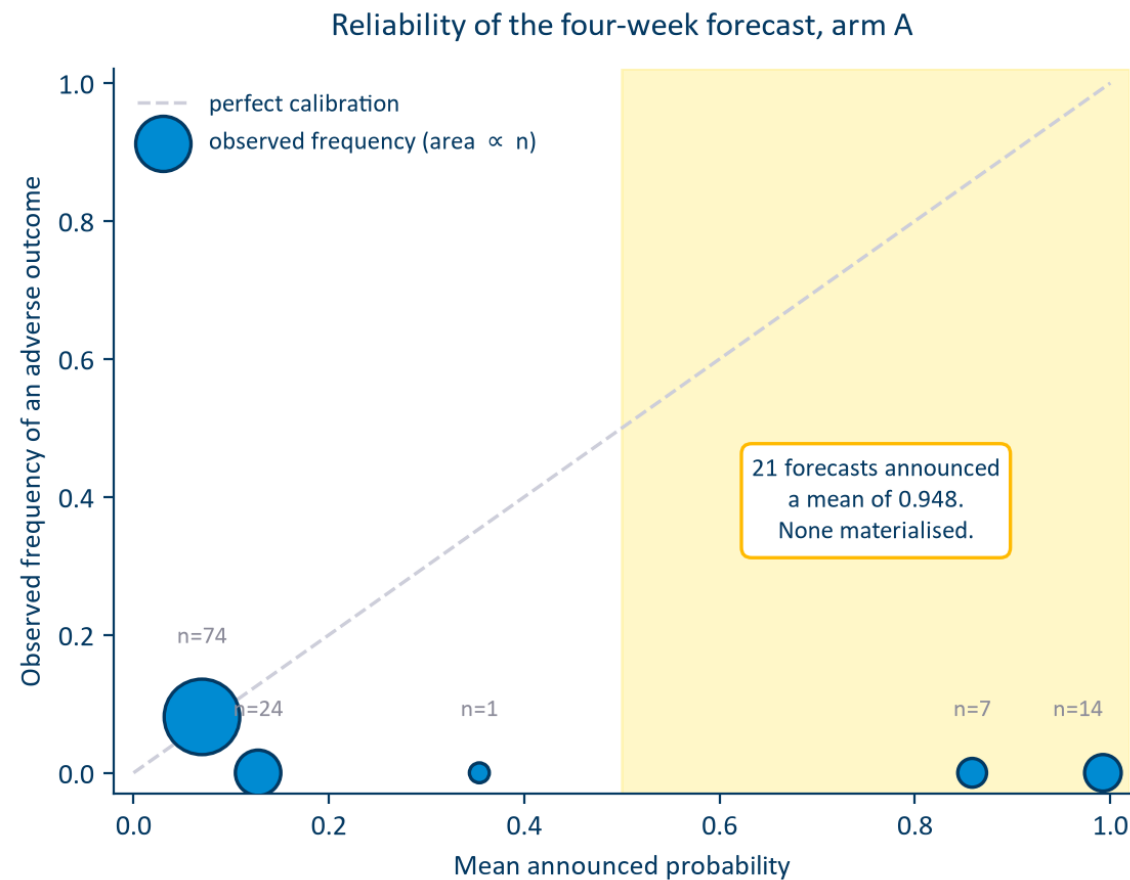
Early warning, not crisis management. Encouraging on eight nodes, not a rate.

THE DASHBOARD THE COORDINATOR READS



RESULT 2: ONE BLOCK EXPLAINED IT

Scored → diagnosed → repaired → re-measured, on the same 120 frozen forecasts



Diagnosed: events never write to the time axis. Across **328 indicator snapshots** the accumulated-delay field was written **zero times**

Repaired: lost time is additive. $completion = t + (1 - progress) \times lead + lost\ time$

BRIER SKILL, BEFORE → AFTER

HORIZON	before	structure	+ uncert.
1 week	-1.278	-0.646	-0.687
2 weeks	-0.690	-0.401	-0.367
3 weeks	-0.654	-0.360	-0.286
4 weeks	-0.553	-0.272	-0.216

The calibration deficit roughly halved at every horizon. **AUC movement is not interpretable at this sample size** : with twenty-six positives the four-week interval runs from 0.21 to 0.87.



Progress now drawn per replicate, widened by each node's own hidden risk, so **the tool calibrates itself**



Indicator coverage rose 25.6 % → 49.1 %. **An empty database hides the model's errors**



Still negative at every horizon. Reported as such. 15 | 20 **ALTEN**

RESULT 3: THE DOMAIN ESTABLISHED

A second chain, and the cheapest thing that could replace the forecast layer

WHICH RANKING FINDS THE THREE FAILING SUPPLIERS?

Adequacy score	0.00
Hidden risk	0.00
Forecast probability	0.56

The three suppliers that miss declare the **1st, 2nd and 4th highest** urgencies from the opening turn. At turn 0 they declare 0.982, 0.941 and 0.876 against computed states of 0.655, 0.838 and 0.542. Hidden risk is computed minus declared, clipped at zero, so nothing selects them against a base rate of 0.20. Their free text says they meant to wait for a recovery plan.

THE MISPERCEPTION BOUNDARY

The tool measures **misperception**, not **non-disclosure**. Precision 1.00 on chain 1 and 0.00 on chain 2 do not contradict: they name the domain.

500 MONTE CARLO TRAJECTORIES VS. A TWO-NUMBER RATIO



6 / 8

horizons where the **baseline ranks better**, by 0.082 of AUC at one week



-1.60

simulation Brier skill vs. **-4.85** for the baseline. The trajectories buy calibration

six urgency blocks



AR(1) state



client-impact channel
displayed, never scored

lead-time draw



milestone term



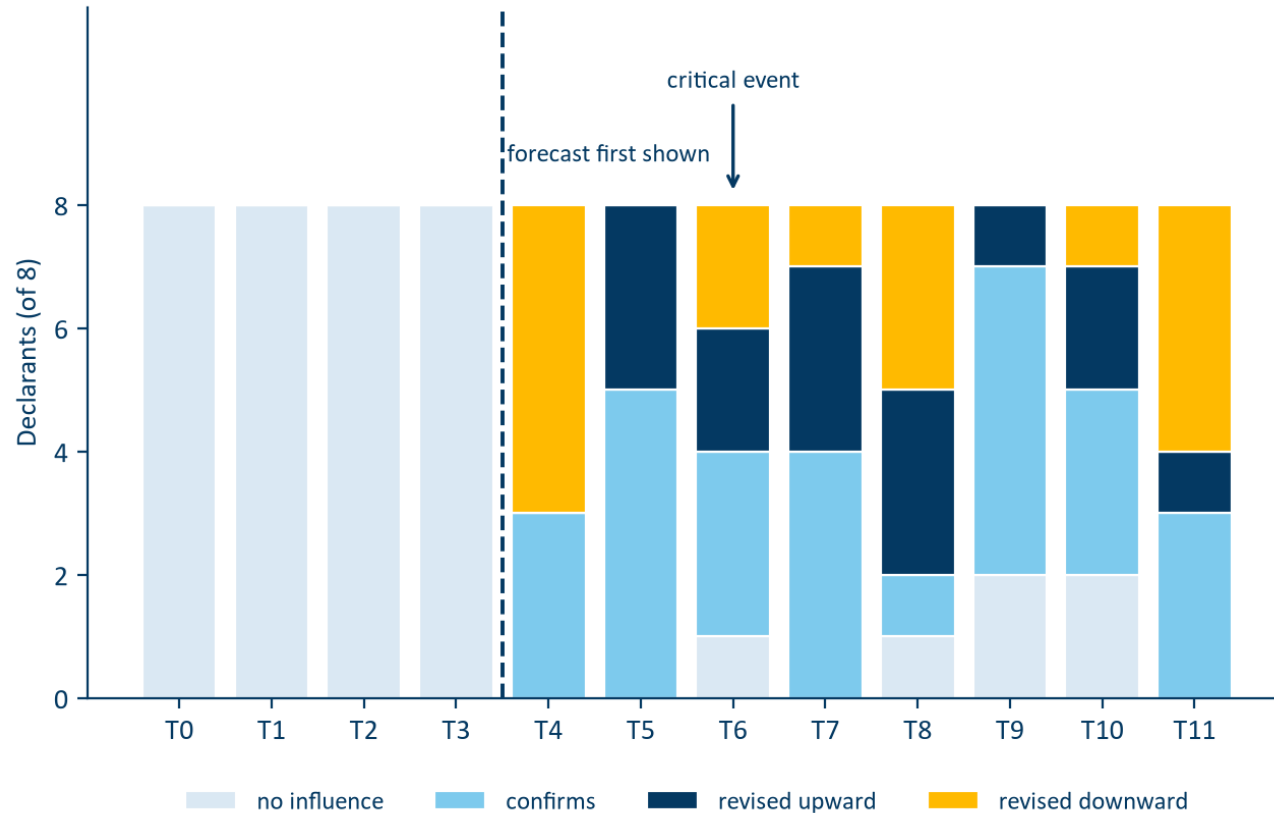
the quantity that is scored

On this chain the forecast layer is a **calibration wrapper around a lead-time ratio**. Defensible, but not what the interface presents. The fix is one line: the AR state must reach the milestone term.

THE RESULT NOBODY PLANNED

Showing the forecast before asking for a declaration changes the declaration

Arm B: recorded influence of the displayed forecast, by turn



5 down - 3 confirmed - 0 up

At first contact all eight declarants record an influence and **not one revises upward**. Reproduced exactly in the repaired run.

15 up - 16 down

Over the full campaign the balance is even. The one-sided effect belongs to **first contact**, shown as a probability with a confidence interval.

1 in 10 - max 0.93

On the second chain, turn-to-turn jumps above 0.20 on unchanged terrain. **Disengagement is the rational response**, not negligence.

OBJECTIVES VS. OUTCOMES

2 campaigns · 2 chains · 23 nodes · 33 milestones · 510 declarations

01	Met	Six blocks, bidirectional propagation, asymmetric score — every term decomposable
02	Met	SupplyScore delivered: multi-tenant, every write audited, tested, documented
03	Met	Propagation and criticality operational; the saturation collapse found and repaired
04	Met, domain established	Right at node level on chain 1; scored, located, repaired, re-measured; then bounded by chain 2 and the baseline
05	Met	Every reported figure regenerated from raw artefacts by script, not transcribed
06	Met, reframed	Docks cannot be counted from nadir, established by measurement. Segmentation reaches –35 % bias

WHAT THE PROGRAMME CAN NOW SAY ABOUT SUPPLYScore



Detects misperception, not non-disclosure



An early-warning tool that closes once a crisis is public



Its forecast buys **calibration**, not short-horizon discrimination

LIMITS · NEXT · WHAT I TAKE

Three limits of sample, one of method, and a list produced by measurement

LIMITS



Agents, not human consultants



Small positive class: 4 of 8, 3 of 15



Carbon block least exercised, for want of node-level data

Methodological, and it does not dissolve with data: no number holds without its outcome definition

FOR ALTEN, IN ORDER

1



Connect computed urgency to the forecast quantity. **One afternoon of work.**

2



Fill the indicators before calibrating anything

3



Make the forecast state its own ignorance; smooth the alarm

4



Run the campaign with humans, the only run that decides

THE LESSON I KEEP

An early dry run derived the declared value from the computed one, and had to be discarded.

I would now design the independence of two signals **before** designing either.

And: before trusting a component, write the cheapest thing that could replace it.

ESTIA COMPETENCY FRAMEWORK

CI1

CI3

CI5

CE3

CE4

CE5

CST1

CST2

CST7

The difficulty was never the mathematics. It sits at the two joins: model to world, and output to the person who acts on it.

THANK YOU

QUESTIONS

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